LITERATURE REVIEW

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Blockchain-Based Academic Degree Verification & Credential Authentication

Class: B.Tech CSE ICBT (R)

Guide:

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1. Introduction

Academic credential fraud is a growing global problem. Diploma mills, forged transcripts, and digitally altered certificates cost economies billions annually and place unqualified individuals in roles where competence is critical — from medicine and engineering to financial services. Traditional verification systems rely on manual registrar queries that can take days to weeks, remain vulnerable to document forgery, and collapse entirely when an institution closes. Even digital PDFs are trivially falsifiable.

This literature review examines how blockchain technology — originally conceived as the infrastructure for Bitcoin (Nakamoto, 2008) — can be applied to create a tamper-proof, decentralised, and instantly verifiable academic credentialing system. Its core properties of immutability, decentralisation, transparency, and cryptographic security map directly onto what a robust credential verification system requires.

Scope: This review covers peer-reviewed articles, government reports, and technical standards published between 2016 and 2025, sourced from IEEE Xplore, ACM Digital Library, Scopus, and the W3C. The central research questions are: (1) What blockchain properties make it suitable for credentialing? (2) What platforms exist and what do they achieve? (3) What are the main technical and governance barriers? (4) What research gaps remain?

2. Review of Existing Systems and Related Work

A growing ecosystem of blockchain credentialing platforms has emerged since 2015. We reviewed five major systems to understand what works, what does not, and what is missing.

Platform	Blockchain	Privacy	Scale	Key Strength
Blockcerts	Bitcoin / Ethereum	Public (hash only)	Global	Open-source; widest institutional uptake
OpenCerts (SG)	Ethereum	Public (hash only)	2M+ certs	Government-mandated; proven at national scale

Sony Global Ed.	Hyperledger Fabric	Private / permissioned	Japan	GDPR-aligned privacy-by-design
EBSI (EU)	Permissioned (EU nodes)	Consortium	27 nations	Cross-border legal framework; W3C VC compliant
EduCTX	Ark Blockchain	Public	Pilot only	First ECTS-compatible blockchain credit system

Blockcerts (MIT Media Lab / Learning Machine): The most widely adopted open-source standard. It anchors a cryptographic hash of the credential into a Bitcoin or Ethereum transaction. Any verifier can independently check the credential against the chain without contacting the institution. Grech and Camilleri (2017) confirmed its technical viability; deployments in Malta, Bahrain, and MIT itself demonstrate institutional feasibility.

OpenCerts (Singapore): The most successful large-scale production deployment. GovTech Singapore issues over two million certificates using Ethereum smart contracts under a government mandate — the critical adoption accelerant (Yong et al., 2021). It is W3C Verifiable Credentials compliant and fully open-source.

EduCTX (Turkanovic et al., 2018): The first platform to replicate the European Credit Transfer System (ECTS) on a blockchain, enabling cross-border credit recognition without a central registry. Piloted with five European universities; now one of the most-cited technical papers in the domain (1,200+ citations).

Sony Global Education / EBSI: Sony uses Hyperledger Fabric (permissioned) for privacy-by-design aligned with Japanese norms and GDPR. The European Blockchain

Services Infrastructure (EBSI) goes further — a consortium network across all 27 EU member states using W3C DIDs and VCs for cross-border diploma recognition, though governance complexity across jurisdictions remains a challenge (Erkkilä & Räsänen, 2023).

3. Research Gaps Identified

3.1 No Universal Interoperability Standard

The proliferation of incompatible credentialing platforms — Blockcerts, OpenCerts, EBSI, EduCTX, and numerous proprietary systems — means an employer cannot verify credentials from different sources with a single workflow. Dib et al. (2020) argue that convergence on the W3C Verifiable Credentials (VC) and Decentralised Identifier (DID) specifications is the single most important determinant of long-term adoption, yet uptake remains fragmented.

3.2 Privacy-Regulation Tension (GDPR vs. Immutability)

GDPR's right to erasure is structurally incompatible with the immutability of public blockchains. Finck (2018) identifies this as the most significant legal barrier. Proposed solutions — storing only hashes on-chain, chameleon hash functions (Ateniese et al., 2017), Zero-Knowledge Proofs — are technically promising but not yet standardised or universally deployed. The legal status of an irreversible hash as 'personal data' remains unresolved across EU jurisdictions.

3.3 Scalability and Key Management

Public blockchains historically process 15–30 transactions per second — far below the volume of global annual degree awards. While Layer-2 solutions and off-chain hybrid architectures address throughput, they introduce new complexity. Separately, Cheng et al. (2022) found that no current approach satisfactorily balances cryptographic key security, usability, and recoverability for non-technical users — a critical gap since losing a private key means losing access to one's credentials.

3.4 Digital Equity and Inclusion

Bouchrika et al. (2021) note that digital-wallet dependency excludes populations without reliable internet or digital literacy — precisely the groups most in need of portable, verifiable credentials (refugees, migrant workers, graduates of closed institutions). No major platform has published a systematic equity impact assessment.

Gap	Research Gap	Impact on Adoption	Proposed Solution

G1	No universal credential standard	Fragmented verification across platforms and borders	W3C DID/VC adoption as baseline standard
G2	GDPR vs. immutability conflict	Legal barrier to public blockchain deployment in EU	Off-chain data; hash-only anchoring on-chain
G3	No longitudinal user studies	Weak empirical evidence for long-term adoption	Multi-year mixed-methods evaluation frameworks
G4	Digital equity gap	Excludes low-connectivity and low-literacy users	Mobile-first, low-bandwidth wallet interfaces

4. Summary of Findings

The literature consistently confirms that blockchain addresses the core failures of traditional credentialing: immutability prevents post-issuance forgery; decentralisation removes single-institution dependency; cryptographic verification eliminates the need for registrar queries. Singapore's OpenCerts deployment demonstrates that national-scale production is achievable when backed by regulatory mandate.

However, the evidence is equally clear that blockchain is a necessary but not sufficient condition for a trustworthy credentialing ecosystem. Governance — standards convergence, institutional change management, and legal alignment — is the critical limiting factor. Mikroyannidis et al. (2020) found that 64% of university administrators viewed blockchain credentialing positively in principle, but only 18% had concrete implementation plans, citing cost uncertainty and regulatory ambiguity.

4.1 Key Dimensions Assessment

Dimension	Rating	Key Insight from Literature
Immutability / Tamper-	Strong	Hash-anchoring on any major chain provides

resistance		effectively permanent records.
Verification Speed	Strong	Instant self-service verification vs. days for manual registrar queries.
Scalability	Moderate	Off-chain hybrid architectures required; Layer-2 solutions still maturing.
Privacy Compliance	Moderate	Hash-only approach works; ZKP selective disclosure still non-standard.
Interoperability	Weak	No universal standard yet; W3C VC/DID adoption is growing but incomplete.
Institutional Adoption	Weak	Low without regulatory mandate; change management underestimated.

5. Conclusion

This literature review affirms that blockchain offers a technically sound foundation for academic degree verification and credential authentication. The convergence of real-world deployments — from MIT Blockcerts to Singapore OpenCerts to the EU's EBSI — demonstrates that the concept has moved well beyond theoretical proposal into operational systems.

The path to widespread adoption runs through three parallel tracks: technical maturation of Zero-Knowledge Proofs and key-recovery mechanisms; regulatory clarity on GDPR compliance and cross-border legal recognition; and institutional change management supported by open, interoperable standards. Research gaps in longitudinal evaluation, user experience, economic modelling, and equity impact must be addressed to build the evidence base that policymakers and institutions need to commit to adoption.

Recommendations: Policymakers should mandate W3C DID/VC standards for publicly funded qualifications. Institutions should pilot open-source platforms to avoid vendor lock-in. Researchers should prioritise mixed-methods, multi-year studies of deployed systems incorporating equity and economic dimensions.

6. References

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